

Task -1

AAA in a Private Wi-Fi Network

Scenario: A student wants to connect their laptop to a **private campus Wi-Fi network** using their university username and password.

Step-by-Step AAA Process

AAA Function	What Happens
1. Authentication	The student enters their campus username and password on the Wi-Fi login page. The authentication server checks the credentials and confirms that the user is a valid campus member.
2. Authorization	After successful authentication, the network determines what the student is allowed to access . For example, the student may access the Internet and university websites but may not access staff-only servers.
3. Accounting	The system records information about the user's network session, such as login time, logout time, username, and amount of data used . Administrators can use these records for monitoring, troubleshooting, and usage management.

Simple Flow

User connects to Campus Wi-Fi

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AUTHENTICATION

"Who are you?"

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AUTHORIZATION

"What can you access?"

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ACCOUNTING

"What did you use?"



Network Access

Conclusion: AAA provides three important controls: **Authentication verifies identity, Authorization controls access, and Accounting records network usage.**

Task -2

AAA Deployment Models Comparison

For a company like **Zomato**, which handles sensitive customer, restaurant, employee, and payment-related information, AAA (Authentication, Authorization, and Accounting) can be deployed in different ways.

AAA

Deployment Advantages Model

Disadvantages

Centralized AAA

1. **Centralized management** – user accounts and access policies can be managed from one location.2. **Consistent security policies** – the same authentication and authorization rules can be applied across systems.3. **Easier auditing** – accounting and login records can be collected centrally.

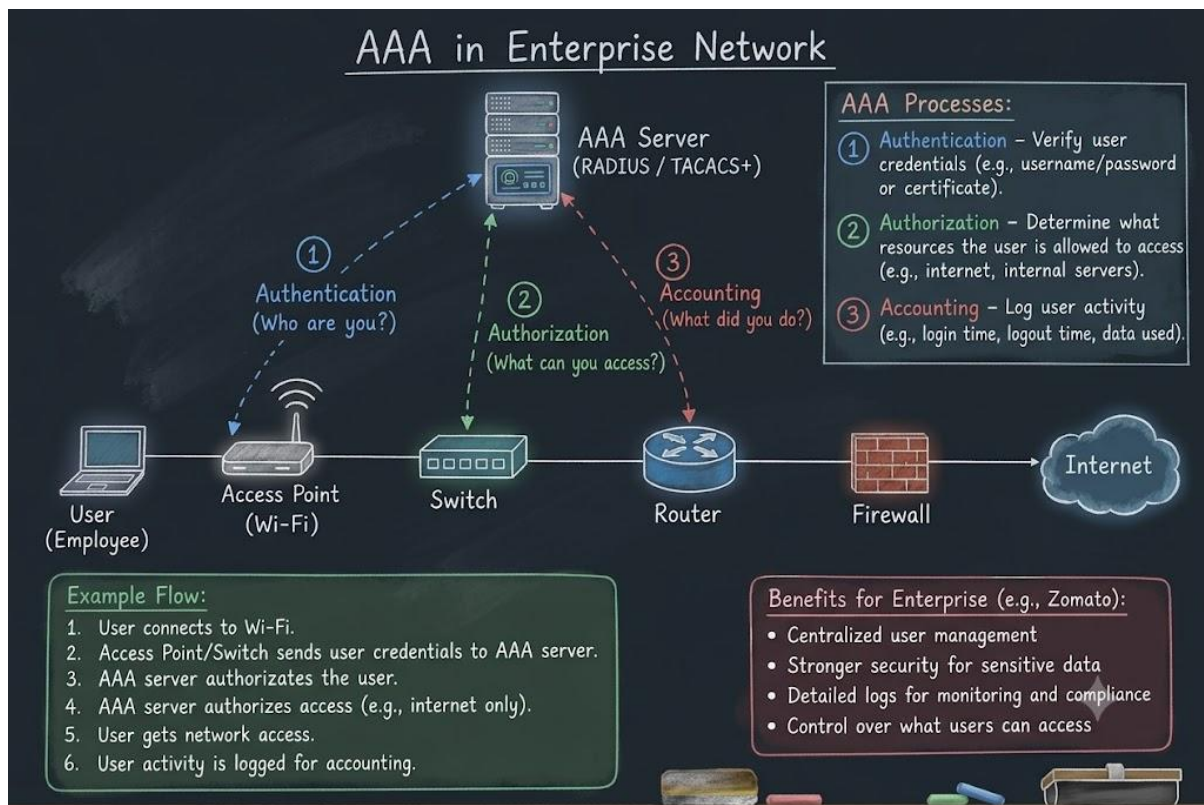
1. **Single point of failure** – if the central AAA service becomes unavailable, authentication may be affected.2. **Scalability challenges** – a large organization needs sufficient capacity to handle authentication requests.3. **Network dependency** – remote systems may have difficulty authenticating if connectivity to the central service is lost.

Distributed AAA

1. **Better availability** – multiple AAA servers/sites can continue operating if one fails.2. **Improved scalability** – authentication requests can be distributed across multiple servers.3. **Local control** – different locations or departments can manage their own authentication requirements.

1. **More complex management** – multiple AAA servers and policies must be maintained.2. **Higher cost** – additional servers, software, and administration may be required.3. **Policy consistency can be difficult** – configurations must be synchronized to avoid different security rules across locations.

Task-3



Task-4

AAA in Paytm

Paytm can use **AAA (Authentication, Authorization, and Accounting)** to secure user accounts and activities. **Authentication** verifies the user's identity when they log in using a mobile number, password, OTP, or biometric authentication. **Authorization** determines what the user is allowed to do, such as sending money, paying bills, viewing transactions, or managing bank accounts. **Accounting** keeps records of activities such as login time, transactions, payment amounts, and device information. These records help detect suspicious activity, provide transaction history, improve security, and support billing and dispute resolution.

Task-5

AAA Policy – College Wi-Fi Network

Purpose:

The college Wi-Fi network will use AAA (Authentication, Authorization, and Accounting) to ensure that only authorized students can access the network and that their network activities are properly controlled and recorded.

1. Authentication:

Students must log in using their college-issued username and password or student ID with OTP. The AAA authentication server will verify the student's identity before allowing network

access. Faculty, staff, and unauthorized users will not be allowed to connect to the student Wi-Fi network.

2. Authorization:

After successful authentication, the system will check the student's access permissions. Students will be allowed to access approved educational websites, college portals, online learning platforms, and other permitted services. Restricted or harmful websites and unauthorized network resources will be blocked.

3. Accounting:

The AAA server will record login and logout times, username/student ID, device information, IP address, and access attempts. Failed login attempts and blocked website requests will also be logged for security monitoring and troubleshooting. Logs will be stored securely and accessed only by authorized college network administrators.

Policy Rules:

- Only currently registered students may use the student Wi-Fi network.
- Each student should use their own account and must not share credentials.
- Educational and college-related websites will be permitted.
- Restricted websites and unauthorized services will be blocked.
- Successful and failed authentication attempts will be logged.
- Suspicious activity may be investigated by authorized network administrators.
- AAA logs will be protected from unauthorized access.